UNIVERSITY OF LEICESTER

CO3102/CO7102
All candidates

Semester 1 Examinations 2025

DO NOT OPEN THE QUESTION PAPER UNTIL INSTRUCTED TO DO SO BY THE CHIEF INVIGILATOR	
School	School of Computing and Mathematical Sciences
Module Code	CO3102/CO7102
Module Title	Mobile and Web Applications
Exam Duration	Two Hours
CHECK YOU HAVE THE CORRECT QUESTION PAPER	
Number of Pages	8
Number of Questions	2
Instructions to Candidates	This paper contains **TWO** questions. Each of the questions is worth 50 marks. Full marks may be obtained for answers to both questions. All necessary namespaces, classes, API helpers and JSTL core tags are provided in the appendices.
FOR THIS EXAM YOU ARE ALLOWED TO USE THE FOLLOWING:	
Calculators	Permitted calculators are the Casio FX83 and FX85 models
Books/Statutes provided by the University	Yes. Any resources found on Blackboard, in textbooks, lecture notes, or other materials accessible online or in print.
Are students permitted to bring their own Books/Statutes/Notes?	Yes. Any books, online or in print. Note: the use of any Generative AI is not permitted in this assessment
Additional Stationery	Yes. Any writing Instruments, paper and computer accessories

1. (a) Explain why a well-formed XML document is not necessarily a valid XML document. [3 marks]

   (b) Consider the XML document `gallery.xml` (see **Appendix A**). Is this document well-formed? Justify your answer. [4 marks]

   (c) Consider the document `places.xml` (see **Appendix B**) representing a family tree. Answer the questions below:

      i. Explain the following XPath expressions, stating the line numbers of any selected elements (or explain why if there is no match rows):

         (1) `//area[@label="Rutland"]/settlement` [2 marks]

         (2) `//settlement/@*` [1 mark]

      ii. For each question below, write an XPath expression to get the specified result:

         (1) Count the number of cities in Leicestershire. [2 marks]

         (2) List the county where Oakham is located. [2 marks]

         (2) List all settlements in East Midland. [2 marks]

      iii. Rewrite lines 8-11 of `places.xml` in JSON format. [3 marks]

      iv. Write an XSD schema that validates against documents like `places.xml`. [7 marks]

   (d) Which of the following statements are true? For each of the false statements, briefly justify your answer.

      i. Arrays and objects are passed by value in JavaScript.

      ii. RESTful API calls do not maintain any state between requests.

      iii. The latest ECMAScript specification replaced reserved keyword `var` with `let`.

      iv. It is more memory-efficient to store login credentials in SQLite database on the mobile devices.

      v. A client can block tracking cookies in the browser to disable the HTTP sessions.

      vi. Progressive web apps have access to native device features such as GPS and camera.

      vii. Node.js is a client-side runtime that built on Chrome's V8 JS engine .

      viii. Android SDK can be used to develop cross-platform apps for iOS.

   [8 marks]

(e) Consider the JavaScript code snippet below:

```
1  let func1=(var1=1,var2="1")=>(var1===var2)
2  var var1=1;
3  var var2=func1;
4  {
5  let var1=2;
6  var2={name1:"func1"};
7  }
8  var var1;
9  let var3=(var2)=>var2*2;
10 var5=func1(typeof(var2), typeof([1,2]))
11 var var4=func1(1);
12 console.log("output="+var1+","+var2+","+var3(var1)+","+var4+","+var5);
```

Answer the following questions:

i. Explain what happens between line 2 and line 9. [4 marks]

ii. Explain the concept of "loosely-typed" in JavaScript with an example from the code above. [3 marks]

iii. Explain the arrow function syntax in line 1 and identify which feature is not supported in ES5. [3 marks]

iv. What is the output of the code above? If an error occurs, explain the cause of the error. [2 marks]

(f) Consider the following HTML snippet:

```
<div class="d1" id="a">
       <p class="s1" id="b">text1</p>
       <p class="s1" id="c">text2</p>
       <a href="#c1">c2</a>
 </div>
```

For each of the question below, write one CSS/jQuery selector statement to match the specific element(s) in the above HTML.

i. Match all elements [1 mark]

ii. Match any hyperlinks enclosed within <div>..</div>. [1 mark]

iii. Match any <p> elements with class name s1. [1 mark]

iv. Match any elements with id="b". [1 mark]

2. (a) Explain the concept of Mobile-First Responsive Web Design. [4 marks]

(b) In general, there are three different approaches for mobile apps development:

- Native Apps (iOS/Android)
- Progressive Web Apps (PWAs)
- Hybrid Apps

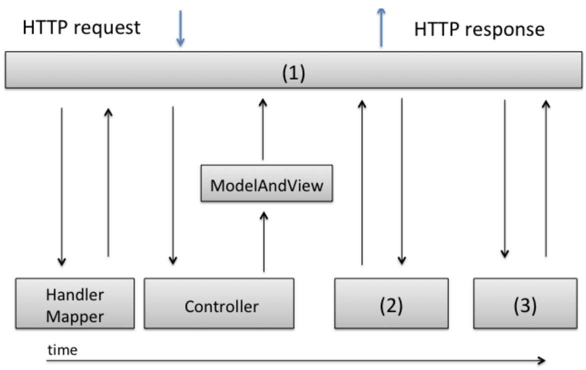
For each of the apps below, which is the most appropriate development strategy? Briefly justify your answers. [12 marks]

i. eBike rider app: a cross-platform app built using web technologies in a single codebase, users and scan the QR code or use NFC-based (Near Field Communication) to pay for their ride.

ii. Tube Planner: Helps users plan their underground journeys using their current location and OpenStreetMap (OSM) . While traveling, users can read downloaded newspapers. Important announcements and live traffic updates are sent via push notifications, which are delivered once the app reconnects to the internet upon reaching a platform.

iii. First-Aid CPR: developed by a charity to provide essential First-aid information. It utilises SEO (Search Engine Optimisation) to make information accessible and ensures its content can be indexed by Google for easy access.

(c) Consider the Spring MVC (Model-View-Controller) flow diagram below:



i. Name the components indicated in the diagram as (1) and (3). [2 marks]

ii. What is the purpose of component in (2). [2 marks]

(d) Which of the following statements are true? (No justification required). [6 marks]

i. In MVVM pattern, a "view model" is another representation of a model for a view.

ii. The `<script>` element must be placed after the `<header>` in an HTML document.

iii. React.js is a library for rendering views. Any updates to the virtual DOM tree will be reflected in the browser.

iv. Although AJAX stands for Asynchronous JavaScript and XML, it also supports JSON and can function without XML or JavaScript.

v. The browser maintains a conversational state with the HTTP server.

vi. Deleting all cookies associated with a website will always log you out of that website.

(e) DVLA (Driver and Vehicle Licensing Agency) provides an online road tax calculator, which is calculated based on a vehicle's CO2 emissions the first time it's registered. The calculator first retrieves the vehicle model from DVLA database using its registration number, then obtains accurate CO2 emission data through another web service offered by ACEA (European Automobile Manufacturers' Association) Vehicle Registry. Your tasks are to complete the code in `DVLA_Controller.java` and `DVLA_RoadTax.html` (the missing part of **the code you have to fill in is indicated by '[...]'**, refer to **Appendix C** for relevant classes and the API helper.)

i. `DVLA_Controller.java` - Complete the URL mappings and implement the method body for the RESTful controller method `getVehicleTaxDetails()`. The requested URL should return a response in the required JSON format.

[6 marks]

ii. `DVLA_RoadTax.html` - Sending AJAX requests to the REST service to fetch vehicle road tax details and display them on this HTML page as the user enters the vehicle registration.

[6 marks]

iii. Provide a brief explanation of how your page requests and process the data to/from the server. [3 marks]

(f) Imagine you are leading new NHS Digital team to develop a mobile app aimed at improving the efficiency in primary care, such as booking GP appointments, accessing and sharing medical records, conducting video consultations, and managing online pharmacy prescription deliveries or collections. Identify the key features you would incorporate into this app, and discuss how each feature could be implemented. Include any technologies, access to hardware features, or external services required to achieve these functionalities. Use a diagram if helpful to illustrate your design.

[9 marks]

Appendix A: gallery.xml

```
<?xml version="1.0" encoding="UTF-8"?>
  <exhibition location="Victoria and Albert Museum">
    <collection>
       The Temptation in the Garden of Eden
    </collection>
    <opening day=Saturday day=Sunday>
    <2025>South Kensington</2025>
    <2024>Dundee</2024>
</opening>
</exhibition>
 <comment>GenAI not allowed :) </comment>
```

Appendix B: places.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<area label="East Midland" type="region">
       <area label="Rutland" type="county">
       <settlement type="town">Oakham</settlement>
       </area>
       <area label="Leicestershire" type="county">
              <settlement type="city">Leicester</settlement>
              <area label="Charnwood" type="district">
                     <settlement type="village">Rothley</settlement>
                     <settlement type="village">Quorn</settlement>
              </area>
       </area>
</area>
```

XML Schema namespace: `xmlns:xs="http://www.w3.org/2001/XMLSchema"`

Appendix C

For **Question 2(e)**: Controller class, HTML webpage and API helper.

(a) `DVLA_Controller.java`

```
@Controller
public class DVLA_Controller{
@Autowired
ACEA_Service ACEA_service;
/*
* GET /DVLA_service/vehicle_detail_by_registration/AB123XY

* Get the vehicle road tax by the registration number e.g. AB123XY
* See Appendix C (b) for the response in JSON
*/
[...]
public [...] getVehicleTaxDetails([...]) {
[...] //Question 2 e(i): complete the code here
}
```

(b) `GET /DVLA_Service/vehicle_tax/AB123XY - an example of JSON response`

```
{
    "vehicle_registration": "AB123XY",
    "model": "BMW X7 xDrive40i 2024",
    "CO2": "232",
    "tax": "410.0"
}
```

(c) `DVLA_RoadTax.html`

```
<html>
 <head><title>DVLA Road Tax checker</title></head>
 <body>
<!-- all required CSS, JS and jQuery libraries are imported... -->

<form aaction="/DVLA_payment">
    <label for="reg">Car Registration e.g. AB123XY </label>
                      <input type="text" id="reg" name="reg"/>
   <label for="tax">Your vehicle's Road Tax for 2024/25 is: </label>
                      <input type="text" id="tax" name="tax" readonly />
   <input type="submit" value="Pay Vehicle Tax" />
</form>

<script>
$(document).ready(function(){
[...] //Q2 e(i): complete the JavaScript code
});
 </script>
</body>
</html>
```

(d) For **Question 2(e)**: Assuming these Service classes have been implemented with the following methods:

```
//ACEA_Service.java (web service provided by external organisation ACEA)
public double getVehicleCO2Emission(String vehicle_model)
//DVLA_Service.java
public TaxDetails getVehicleTaxByEmission
                         (String vehicle_model, double CO2emission)
```

- The method `getVehicleCO2Emission` returns a specific vehicle model's CO2 emission data (g/km).
- The method `getVehicleTaxByEmission` returns the vehicle road tax as a `TaxDetails` object.

(e) For **Question 2(e)**:

```
class TaxDetails{
       String vehicle_registration;
       String model;
       String CO2;
       double tax;
}
//assuming all getters/setters are provided.
```